

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 3.2: Growth and Development in Animals — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 3.2: Growth and Development in Animals
Driving Question	DRIVING QUESTION: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Empty Skin on the Maize Leaf: Two Ways Insects Grow	Learners examine real or pictured specimens from a Kenyan maize field: an empty grasshopper skin (exuvia) on a leaf, grasshopper nymphs of different sizes, and the life stages of a butterfly or the maize stalk borer (egg, larva, pupa, adult). Teacher note: use WAIT TIME of 10 seconds after asking 'What do you think the empty skin is?' and cold-call 3 learners to share predictions before revealing the answer. Learners notice that grasshopper nymphs look like small adults, whereas caterpillars and maggots look nothing like the adult.	Insects follow two developmental pathways. Incomplete metamorphosis (egg, nymph, adult: 3 stages), as in the grasshopper, has nymphs that resemble small wingless adults and grow through successive moults. Complete metamorphosis (egg, larva, pupa, adult: 4 stages), as in the butterfly, housefly and maize stalk borer, has a larva that looks totally different from the adult, and a pupa stage in which the body is reorganised. Teacher note: stress that the empty skin is the shed exoskeleton, evidence that the insect grew by moulting.	The empty skin on the maize leaf is the cast-off exoskeleton left when a grasshopper nymph moulted and grew larger. Model answer: a grasshopper grows by shedding its skin several times, and each nymph stage looks like a smaller adult (incomplete metamorphosis), while a butterfly or stalk borer changes shape radically through larva and pupa stages (complete metamorphosis). Common misconception to correct: the empty skin is not a dead insect. Add the first questions to the DQB: 'Why must the insect shed its skin?' and 'Why does a caterpillar not look like a butterfly?'
Lesson 2 Stepwise Growth: Why Hard-Shelled Animals Grow in Jumps	Learners study a table or graph of body length of a grasshopper nymph (or a crab from the Mombasa coast) measured over several weeks, and plot size against time. They see flat periods when the size stays the same, followed by sudden jumps. Teacher note: ask learners to 'describe the shape of the graph' before interpreting it, allow 10 seconds of WAIT TIME, and cold-call 3 learners. A simple model with a tight-	Animals with a hard exoskeleton cannot grow smoothly because the shell does not stretch. They must moult (ecdysis) first: the old cuticle is shed, then the animal swells by taking in air or water while the new cuticle is still soft, and then the new cuticle hardens (sclerotisation). Growth therefore happens in jumps (a stepwise growth curve), with the periods between moults, called instars, showing almost no increase	Model answer to the driving question: an insect outgrows its hard shell by moulting, shedding the old exoskeleton, expanding quickly while the new one is soft, then hardening it. This produces a staircase growth curve rather than a smooth one. Vulnerability point: a freshly moulted insect is soft and exposed to predators such as birds, so moulting is risky. Update the DQB with: 'What tells the insect when to moult?'

	fitting paper sleeve or a non-stretching glove helps show why a rigid covering limits growth.	in size. Teacher note: emphasise that tissue growth can occur inside, but external size increases only at moulting.	Learners add the stepwise curve to the class model.
Lesson 3 Growth Curves: Why Do Vertebrates and Soft-Bodied Animals Grow Smoothly in an S-Shape While Insects with Hard Shells Grow in Steps?	Learners plot and compare data sets: the mass of a Kenyan Boran calf or a human child (or tilapia from Lake Victoria fish ponds) over time, and the stepwise length of an insect nymph. Teacher note: give learners graph paper, let them plot individually for 5 minutes, then compare in pairs. They should observe that the vertebrate curve is a smooth S-shape while the insect curve is stepped. Ask: 'Where is growth fastest? Where does it slow down?' and allow WAIT TIME before cold-calling 3 learners.	Vertebrates and soft-bodied animals grow continuously in a sigmoid (S-shaped) curve with four phases: lag (slow start while the organism adapts), exponential or log (rapid growth), deceleration (growth slows as limits such as food and space are reached) and stationary (growth levels off). Growth rate = (final size – initial size) ÷ time taken. Teacher note: work a calculation, for example a calf growing from 30 kg to 150 kg in 6 months gives $(150 - 30) \div 6 = 20$ kg per month. Insects with a rigid exoskeleton show a stepped curve, although their overall trend over time is still roughly sigmoid.	Model answer: vertebrates have an internal skeleton (endoskeleton) that grows with the body, so they increase in size continuously and give a smooth S-curve. Insects have an external skeleton that cannot stretch, so their size increases only at moulting, giving a staircase curve. Learners should be able to label the four phases and calculate growth rate correctly with units. Common error: confusing the stationary phase with death. Add to the DQB: 'What controls the timing of the steps?' and revise the model to show both curve types.
Lesson 4 Hormones as the Control Switches of Moulting and Metamorphosis	Learners study a flow diagram and data from classic experiments (for example Wigglesworth's work on the blood-sucking bug Rhodnius, where decapitation at different times affected moulting) and discuss what happens when the head, brain or glands are removed or hormones are injected. Teacher note: use a 'think-pair-share' with 10 seconds of WAIT TIME and cold-call 3 pairs to state the cause-and-effect chain. Ask: 'What does the evidence tell us about where the moulting signal comes from?'	Brain Hormone (from neurosecretory cells in the brain) signals the prothoracic glands to release Ecdysone, which triggers moulting. Juvenile Hormone (JH), from the corpora allata, decides the kind of moult: high levels keep the insect juvenile (the moult gives another nymph or larva), while falling levels allow the next stage to develop towards the pupa and adult. Teacher note: Ecdysone says 'moult now'; JH says 'what to become'. Sequence: brain → Brain Hormone → prothoracic glands → Ecdysone → moult.	Model answer: moulting is not random but under hormonal control, with Brain Hormone acting as the first switch, Ecdysone as the moulting trigger and Juvenile Hormone as the decision-maker for the type of moult. This answers the DQB question 'What tells the insect when to moult?' Learners should state that endocrine control ensures growth stages happen in the correct order and timing. Add the hormone pathway to the class model and pose a new DQB question: 'How does JH decide between staying young and becoming an adult?'
Lesson 5 The Chemical Switch: How Juvenile Hormone Decides Whether an Insect Stays Young or Becomes an Adult	Learners analyse data tables or diagrams showing JH levels at each moult in an insect, and results of experiments in which extra JH is applied to a larva or JH-producing glands are removed. They observe that extra JH produces an extra larval stage or giant larvae, while removing the glands leads to premature pupation or tiny adults. Teacher note: ask learners to predict before revealing results, allow 10 seconds of WAIT TIME, and cold-call 3 learners to justify their predictions with evidence.	Moulting is triggered by ecdysone, and the level of Juvenile Hormone decides what type of moult occurs: high JH means the insect moults into another nymph or larva (stays juvenile); low JH means it moults into a pupa (in complete metamorphosis) or directly into an adult (in incomplete metamorphosis); no JH at the last moult gives the adult. Teacher note: use the analogy of a 'stay young' dial. Application: JH-mimicking insecticides can stop pests such as mosquitoes from becoming adults, and farmers can use this in integrated pest	Model answer: the same ecdysone signal produces different results depending on the JH level, so JH acts as the chemical switch between staying young and becoming an adult. This explains why a caterpillar grows through several larval moults before pupating. Check understanding with the question: 'What would happen if a locust nymph were given extra JH?' Expected answer: it would stay in a nymph stage and delay becoming an adult. Update the DQB and the model with the JH decision step.

		management.	
Lesson 6 Final Explanation: Controlled Growth, Metamorphosis and Survival in Kenyan Insects	Learners revisit the anchoring phenomenon (the empty skin on the maize leaf) and review their DQB, stepwise graphs and hormone model. They apply their understanding to Kenyan insects such as the desert locust (incomplete), the maize stalk borer, the silkworm and the mosquito (complete), and consider survival advantages. Teacher note: allow WAIT TIME of 10 seconds after each prompt, cold-call 3 learners per question, and have learners present gallery-walk models using evidence from all lessons.	An insect's hard exoskeleton cannot stretch, so the insect moults (ecdysis) under hormone control: ecdysone triggers moulting and juvenile hormone decides whether the next stage is another juvenile or an adult. Incomplete metamorphosis (egg, nymph, adult) gives gradual change, while complete metamorphosis (egg, larva, pupa, adult) gives radical change. Complete metamorphosis reduces competition because larvae and adults use different foods and habitats (for example, caterpillars eat leaves while adult butterflies drink nectar). Vertebrates grow continuously in an S-shaped curve because their skeleton is internal.	Model answer to the driving question: an animal outgrows its hard shell by moulting, shedding the old exoskeleton and expanding before the new one hardens, with growth in steps controlled by ecdysone and JH. Some animals radically change shape (complete metamorphosis) because JH levels fall to allow a pupal stage and adult reorganisation, with survival benefits such as less competition between young and adults, while others, such as grasshoppers and vertebrates, simply grow bigger. Assess using the final model: look for correct use of ecdysis, ecdysone, JH, instars, the stepped versus S-shaped curves and a Kenyan example. Common misconception to resolve: hormones do not make the shell grow; they trigger and direct moulting. Close by resolving the DQB questions, marking each as answered.

ATTRIBUTION AND LICENSING

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